SOFIA ABIDI

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Education

Veermata Jijabai Technological Institute (VJTI), Matunga — **Aug 2024 – Present**
B.Tech in Computer Engineering — CGPA: **9.69/10**

Mukesh Patel School of Technology Management & Engineering, Mumbai — **Jul 2023 – May 2024**
B.Tech in Information Technology — CGPA: **3.98/4.0**

Universal High School, Dahisar — **Aug 2021 – May 2023**
HSC: **94.5%**

Relevant Coursework

- Data Structures & Algorithms
- Algorithm Analysis
- Linux Administration
- Linear Algebra
- Deep Learning / Neural Networks
- Computer Architecture
- Open Source Technology
- Operating Systems
- Reinforcement Learning

Projects

Exoplanet Detection Pipeline | *XGBoost, SHAP, TLS/BLS, Wotan* — **Jun 2026**

- Built, for **ISRO's Bharatiya Antariksh Hackathon (BAH) 2026**, an AI-based pipeline that scanned **200+ TESS light curves** and flagged **22 transit candidates** from noisy astronomical data.
- Applied strong **problem-solving** skills to design an **18-feature** vetting stack (BLS/TLS, odd/even depth, secondary eclipse, centroid shift) feeding a two-stage **XGBoost** classifier reaching **79%/83%** held-out accuracy.
- Added **block-bootstrap** confidence intervals and **SHAP-based** explainability, beating a naive baseline by **+9 macro-F1** across **5-fold CV**.

Parichay: e-Office KPI & Workforce Intelligence System | *React, Node.js, scikit-learn* — **Sep 2025 – Dec 2025**

- Engineered a full-stack **workforce intelligence platform** with role-specific **KPI dashboards** for **Ministry of Jal Shakti** and **Brahmaputra Board**, enabling real-time productivity tracking across **cross-functional teams**.
- Architected **ML pipelines** for performance scoring, workload prediction, staffing optimization, and **anomaly detection** using Python and scikit-learn; solution **presented to the Chairman of Brahmaputra Board**.

TraffIQ | *Python, PyTorch, SUMO, Traci, Multi-Agent Systems* — **Jul 2025 – Oct 2025**

- Designed an **adaptive traffic signal control** system in SUMO using **single-agent and multi-agent reinforcement learning (MARL)**, implementing Q-Learning, DQN, PPO, MAPPO, and Genetic Algorithms.
- **Tested and validated** signal-control policies across multiple traffic scenarios, achieving a **25% reduction in queue length** and **55% decrease in waiting time**, outperforming rule-based baselines.

Achievements

- **Winner, Smart India Hackathon (SIH) 2025**.
- Selected Participant, **BNY Spectrum Joint Certification Program** (Mar 2026): trained in **software architecture & design patterns, clean code practices, and database systems** (SQL/NoSQL) alongside BNY's engineering team.
- Completed **Hacktoberfest 2025** with **10 merged pull requests**, demonstrating strong proficiency in **Git/GitHub workflows**, issue tracking, and branching strategies.
- Placed **2nd** out of 50 teams in **CodeHunt**, a competitive programming contest by Community Of Coders (CoC), VJTI.

Technical Skills

- **Languages:** C++, Python, Java, JavaScript, TypeScript, HTML5, CSS
- **Frameworks & Libraries:** Spring, PyTorch, TensorFlow, scikit-learn, Flask, FastAPI, React, Node.js, OpenCV, MediaPipe
- **Databases:** MongoDB, PostgreSQL, SQL/NoSQL
- **Tools & Platforms:** GitLab, GitHub, Jupyter Notebook, Postman, Agile/Scrum, SDLC

Responsibilities & Volunteering

STEM Mentor | *Agastya Foundation in collaboration with JP Morgan* — **Dec 2025 – Mar 2026**

- Mentored underprivileged school students in designing and building an **IoT-based hardware Decibel Meter**, introducing them to **embedded systems, sensors, and hands-on electronics**.

Marketing Executive | *Institution of Engineering and Technology (IET), MPSTME* — **Jul 2023 – May 2024**

- Secured **sponsorships and brand partnerships** for technical fests; liaised with industry stakeholders to negotiate deliverables.

Social Impact Executive | *Student Council, MPSTME* — **Jul 2023 – May 2024**

- Led large-scale community initiatives including an **Annual Blood Donation Drive** and **Beach Clean-up Campaigns**, coordinating **200+ student volunteers** in collaboration with NGOs.